CUSTOMER AND COMMERCIAL DILIGENCE REVIEW - RECORD 003

Helioscope Robotics, Inc. - diligence file reference CUSTOMER_DILIGENCE-003

Pilot criteria and outcome at Pennarth Industrial

The Wrexham site shows that the charter predates go-live, which is better governance than most accounts show, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. Pennarth Industrial management reads the same way: two criteria were written without a number and were argued about afterwards. The Wrexham site is scheduled at record level rather than in summary, because an account-level average at Pennarth Industrial conceals the site-to-site variation that decides how this relationship behaves at renewal. Ellesmere Port (PEN-02) shows traffic deadlock at an aisle intersection as the complaint theme, on 26 tickets in 2023-Q1. Where the plan assumed a clean run, Wrexham instead took 34 extra days over integration test in the customer sandbox, for supplier lead time on the conveyor tie-in guard package. Wrexham produced 2 severity-one events in 2022-Q4, and resolution attainment of 87 per cent triggered credits of $2,873. Where the plan assumed a clean run, Ellesmere Port instead took 5 extra days over conditional acceptance certificate, for charger electrical permit issued late by the local authority. Ellesmere Port (PEN-02) is where pick confirmation duplicated in the host WMS recurs, and the quarter to 2021-Q3 shows 0 severity-one events. Acceptance slipped 52 days at Ellesmere Port before wms interface build and message mapping was recorded as closed. Recurring at Ellesmere Port through that quarter was lift-plate calibration drift, across 31 tickets. Deployment records at Ellesmere Port leave acceptance test protocol execution closing on 2022-04-01, status closed.

Fieldwork at Pennarth Industrial carries two qualifications: targets are as written in the charter and were restated once, and a criterion measured by manual observation is weaker evidence than one from the host. The Pennarth Industrial bundle also observed, when the schedule was put to the customer, that the mixed result is more credible than a clean sweep would have been, which is consistent with what the documents show without being proved by them. Grainne Mulqueeny's team accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Reviewing the 2022-Q4 quarter, Ellesmere Port shows 23 tickets and a recurring theme of fleet console latency during shift change. Wrexham logged conditional acceptance certificate on 2021-12-24; the customer attributes the slip to supplier lead time on the conveyor tie-in guard package. Recurring at Ellesmere Port through that quarter was pick confirmation duplicated in the host WMS, across 39 tickets. Turning to Ellesmere Port, the tracker puts integration test in the customer sandbox at 2021-11-26 where the plan said 2021-09-20, the gap being labour agreement consultation with the site works council. Ellesmere Port triggered credits of $0 following resolution attainment of 92 per cent. Turning to Ellesmere Port, the tracker puts pilot cell go-live at 2022-02-13 where the plan said 2021-12-08, the gap being wireless survey failed in the high-bay aisles, access points re-sited. Attainment figures from Ellesmere Port show 99 per cent on response against 70 per cent on resolution. Ellesmere Port lost time because seasonal peak freeze: no cutovers permitted during the retail quarter, and site survey and slotting scan moved to 2021-04-19 from a planned 2021-02-14.

Schedule 3.1 - Pennarth Industrial: pilot success criteria and measured outcome

Criterion	Target	Measured	Met	Measurement source	Note
Interface message success rate to the host WMS	88	97	met	fleet console telemetry	criterion agreed in writing before the pilot
Operator induction time to competence	83	87	met	joint stopwatch study	target restated once the baseline was corrected
Charge availability at shift change	83	92	met	fleet console telemetry	target restated once the baseline was corrected

Buying centre and procurement dynamics at Pennarth Industrial

Grainne Mulqueeny, Head of Capital Sourcing controls the paper and Teodor Vaszary, Plant Director controls the budget, and in discrete manufacturing, hydraulic assemblies that split usually decides how a renewal is priced rather than whether it happens. Each was interviewed without the other present, which is how the divergence in their accounts of the

overage dispute came to light. The disagreement was undocumented internally, which suggests the account team is managing two conversations rather than one relationship. Pennarth Industrial's ledger for March shows overage of 17,607 picks, credits of $0 and a variance of $182, no exception raised. Nkechi Adaeze-Fowler was reached by on-site walkthrough and gave a re-buy answer of yes, and has already sponsored an expansion, which sits oddly beside the praise that safety record was clean across two shifts and no lost-time incidents. Billing for December charged Pennarth Industrial $61,402 on a metered volume of 157,762, and the exception was no exception raised. Clementine Ashdown was reached by telephone and gave a re-buy answer of yes, without qualification, which sits oddly beside the praise that firmware rollbacks were quick when a release misbehaved. Console records for July at Pennarth Industrial show 97,127 picks, and the invoice followed at $49,920. Because Lise Vandermolen sees the fleet every shift, the first question put was about inventory record accuracy after cycle counting, answered at 9 per cent. Reconciling February leaves Pennarth Industrial with a variance of $-277, credited in the following billing cycle. Talking to Lise Vandermolen for 30 minutes by on-site walkthrough produced the claim that inventory record accuracy after cycle counting improved by 9 per cent, unverifiable: the customer declined to release underlying data. Pennarth Industrial was credited in January with $496, against an invoiced amount of $49,920. Clementine Ashdown, introduced by Pennarth Industrial as a willing reference, nonetheless complained that traffic management degrades when aisle congestion exceeds design load. Reconciling July leaves Pennarth Industrial with a variance of $-56, no exception raised.

Signature authority sits above both contacts for any increase in committed spend, and the threshold was described to us in terms of a capital approval process rather than a number. Growth here is gated by capital allocation, and the current queue at this customer is dominated by a conveyor retrofit and a transport-management system replacement. The competing projects are not evidence of dissatisfaction, which is worth saying because a flat renewal at this account will look like stalled expansion in a cohort chart. Pennarth Industrial's position for October was settled as no exception raised, on an invoice of $61,552. Nkechi Adaeze-Fowler, speaking as site general manager, said that the roadmap commitments given during the sales cycle slipped a year and pressed the point twice before the call ended. Pennarth Industrial in March shows entitlement of 113,599 against consumption of 131,206, recorded as no exception raised. Obinna Chukwuemeka volunteered that implementation engineers were candid about what the pilot would not prove, and the claim on cost per outbound line was substantiated: the site produced labour reports for the comparison periods. Overage in August at Pennarth Industrial came to 10,981 picks, entitlement having been fixed at signature. Della Fontanez-Ruck answered on re-buy with yes, but only for high-velocity zones and would not be drawn further on price or on term. Pennarth Industrial used in May only 134,520 of an entitled 113,599, and no credit was sought in that cycle. Petra Lindenbaum-Cruz, the distribution centre operations manager, spoke on 2026-03-05 and praised that the WMS interface behaved predictably once message mapping settled, answering the re-buy question with no comment; referred the question to procurement. Pennarth Industrial was credited in June with $0, against an invoiced amount of $49,920. Berenice Straatman, speaking as IT applications manager for the host WMS, said that the roadmap commitments given during the sales cycle slipped a year and pressed the point twice before the call ended. Pennarth Industrial's ledger for August shows overage of 0 picks, credits of $5,700 and a variance of $277, escalated to the account executive, unresolved at fieldwork close.

Procurement's stated intent is to benchmark, and the benchmarking it described is real: a specification was drafted and two vendors were approached informally. No formal tender has issued, and the sponsor said he would resist one on the grounds that a re-integration would cost more than any discount obtained. The sponsor position is the stronger card here, and the account plan we were shown does not mention it. Pennarth Industrial's ledger for November shows overage of 6,628 picks, credits of $0 and a variance of $0, no exception raised. Talking to Lise Vandermolen for 30 minutes by on-site walkthrough produced the claim that inventory record accuracy after cycle counting improved by 9 per cent, unverifiable: the customer declined to release underlying data. Pennarth Industrial was credited in September with $0, against an invoiced amount of $49,920. Nkechi Adaeze-Fowler raised the complaint that the roadmap commitments given during the sales cycle slipped a year, and put the improvement in inventory record accuracy after cycle counting at 20 per cent. Pennarth Industrial was credited in September with $0, against an invoiced amount of $51,790. Nkechi Adaeze-Fowler, speaking as site general manager, said that the roadmap commitments given during the sales cycle slipped a year and pressed the point twice before the call ended. Pennarth Industrial carried overage in May of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Berenice Straatman treats the relationship as one where the fleet-orchestration console gave supervisors usable congestion data, and the cost per outbound line figure offered was 21 per cent. Credits reaching Pennarth Industrial in the May cycle came to $0, and the balance remains as

written off as a goodwill concession. Idunn Haraldsen volunteered that robots tolerated the mixed pallet and tote presentations on the floor, and the claim on reduction in temporary labour headcount at peak was not substantiated: the interviewee described it as an impression. Overage in July at Pennarth Industrial came to 567 picks, entitlement having been fixed at signature.

Related-party and affiliation screening in the Pennarth Industrial review

Affiliation screening was run across the customer base rather than only on the account under review, because a single affiliated customer changes how an investor reads a reference pack. The screen identified one affiliation: Solstice Renewables Field Services is an affiliate of an existing shareholder, contracted on 2022-12-05 with an ACV band of 0.3m to 0.5m. Pennarth Industrial itself showed no affiliation on the register searches performed, and its procurement contact confirmed no ownership or advisory relationship. Punch items numbering 0 remained at Ellesmere Port when integration test in the customer sandbox was signed on 2021-11-26. Lindenvale Grocery renews on 2028-02-14, and Aleksy Wisniewski owns that conversation on the company side. Punch items numbering 0 remained at Ellesmere Port when wms interface build and message mapping was signed on 2021-08-13. Palo Verde Electronics is described internally as warm, technically demanding, low commercial friction, and the movement expected at renewal is flat renewal expected at current fleet size. Certificates held at Wrexham put network and radio-frequency readiness review on 2021-04-12 and explain the 63 days by customer master-data cleanup for slot dimensions ran long. Vantageport Terminals pays on net sixty, a longer cycle than most of the base and one negotiated at signature. Progress at Wrexham shows integration test in the customer sandbox signed off on 2021-07-27 against 2021-06-23 as planned. Comparison with Thornquist Parcel is instructive, since it prices on per-pick metering with a floor commitment and pays on net thirty, monthly in arrears. Where the plan assumed a clean run, Wrexham instead took 14 extra days over conditional acceptance certificate, for supplier lead time on the conveyor tie-in guard package. Matthias Ehrenwerth sponsors the Grindelwald Precision deployment, which covers 1 sites in precision tooling and metrology equipment. Once Wrexham had dealt with wireless survey failed in the high-bay aisles, access points re-sited, the site completed final acceptance certificate on 2022-03-21.

Solstice Renewables was reviewed on its own terms to see whether the affiliation shows in the commercial substance, which is the question that matters rather than the label. The rate compares favourably to peer accounts, and the order form contains a most-favoured-rate statement that no other reviewed contract carries. Its implementation reads like any other account, with slippage and punch items in line with unaffiliated sites, so the relationship is not a paper arrangement. Ignatius Wrenfield sponsors the Sablecreek Apparel deployment, which covers 3 sites in apparel wholesale and returns processing. Wrexham (PEN-01) cleared acceptance test protocol execution on 2021-11-04 against a plan of 2021-10-17, a difference of 18 days recorded against facility readiness: mezzanine steel and charger bays incomplete. The nearest comparable on fleet density is Solstice Renewables, priced on per-robot subscription at a stated most-favoured rate. Wrexham (PEN-01) met punch-list closure review on 2022-02-06 without a variation to plan. Looking across at Ironbark Building Materials, the renewal structure reads as auto-renew annually; one site non-renewed. Punch items numbering 9 remained at Ellesmere Port when conditional acceptance certificate was signed on 2022-05-05. The nearest comparable on fleet density is Grindelwald Precision, priced on per-robot subscription with an on-site engineer allocation. On the evidence from Wrexham, acceptance test protocol execution slipped 18 days and the status now reads closed. Zephyrine Beauty Brands deploys 52 robots across 2 sites and is banded at 0.6m to 0.8m. Once Ellesmere Port had dealt with customer middleware team could not release the interface test window, the site completed punch-list closure review on 2022-07-27. Grindelwald Precision is described internally as new, early and enthusiastic, no renewal history, and the movement expected at renewal is expansion order for one additional site under evaluation.

Two consequences follow, and only the second is significant. The account is small enough that exclusion barely matters, so a reader who prefers to strip related-party revenue out of a growth or retention statistic can do so without the picture changing. What needs legal attention is the rate clause, since a clause negotiated with an affiliate may constrain pricing to unaffiliated customers, and that question has been passed to the legal workstream with the document reference. Caldera Cold Chain is described internally as cool, unresolved reliability grievance in freezer aisles, and the movement expected at renewal is downgrade under discussion to a single-shift tier. Wrexham lost time because customer sponsor on medical leave, sign-off authority unclear, and pilot cell go-live moved to 2021-09-03 from a planned 2021-08-18. Grindelwald Precision is described internally as new, early and enthusiastic, no renewal history, and the movement expected at renewal is expansion order for one additional site under evaluation. Acceptance slipped 5 days at Ellesmere

Port before conditional acceptance certificate was recorded as conditional acceptance issued. Zephyrine Beauty Brands deploys 52 robots across 2 sites and is banded at 0.6m to 0.8m. Certificates held at Ellesmere Port put site survey and slotting scan on 2021-04-19 and explain the 64 days by seasonal peak freeze: no cutovers permitted during the retail quarter. Palo Verde Electronics is described internally as warm, technically demanding, low commercial friction, and the movement expected at renewal is flat renewal expected at current fleet size. Ellesmere Port (PEN-02) met final acceptance certificate on 2022-08-12 without a variation to plan. Ironbark Building Materials began on 2021-10-25 and its structure is described as auto-renew annually; one site non-renewed. Certificates held at Wrexham put network and radio-frequency readiness review on 2021-04-12 and explain the 63 days by customer master-data cleanup for slot dimensions ran long. Northriding Automotive deploys 246 robots across 5 sites and is banded at 1.9m to 2.3m.

Pennarth Industrial: contract shape and account profile

Contracting with Pennarth Industrial Components plc began on 2020-09-01 under an initial term of 60 months, structured as evergreen after initial term, one-hundred-eighty-day notice, and priced as per-robot subscription, tiered by shift coverage. The account bills in the band 0.7m to 0.9m, on net thirty, which places it in the upper half of the customer base by size without making it dominant. Ellesmere Port carries the densest fleet at 41 robots and is the site every reference conversation returns to, favourably or otherwise. Punch items numbering 0 remained at Ellesmere Port when wms interface build and message mapping was signed on 2021-08-13. Asked what worked, Berenice Straatman said that the fleet-orchestration console gave supervisors usable congestion data; asked what did not, the answer was that the roadmap commitments given during the sales cycle slipped a year. Acceptance slipped 67 days at Ellesmere Port before integration test in the customer sandbox was recorded as closed. Asked what worked, Della Fontanez-Ruck said that the account team escalated without being chased; asked what did not, the answer was that the analytics module is priced separately and was never budgeted. Where the plan assumed a clean run, Wrexham instead took 22 extra days over final acceptance certificate, for wireless survey failed in the high-bay aisles, access points re-sited. Talking to Lise Vandermolen for 30 minutes by on-site walkthrough produced the claim that inventory record accuracy after cycle counting improved by 9 per cent, unverifiable: the customer declined to release underlying data. Wrexham is one of the few sites where punch-list closure review ran to plan, closing on 2022-02-06. Clementine Ashdown gave 27 minutes on 2026-03-13 and made one quantified claim, on dock-to-stock hours on inbound receipts, of 23 per cent. Turning to Ellesmere Port, the tracker puts integration test in the customer sandbox at 2021-11-26 where the plan said 2021-09-20, the gap being labour agreement consultation with the site works council. Berenice Straatman would re-buy on the basis of yes, subject to a commercial reset on overage, having first complained that the roadmap commitments given during the sales cycle slipped a year. Punch items numbering 0 remained at Wrexham when network and radio-frequency readiness review was signed on 2021-04-12.

Grainne Mulqueeny, Head of Capital Sourcing owns the commercial relationship, and Teodor Vaszary, Plant Director sponsors it operationally, a split that matters because the two functions do not describe the relationship in the same terms. The operating view is warmer than the commercial view, and management's own characterisation of the account as stable and quietly satisfied, low engagement sits closer to the operating view than to what procurement told us. The contract next turns on 2027-09-01, with notice due by 2027-03-05 under a 180-day window. Della Fontanez-Ruck volunteered that the account team escalated without being chased, and the claim on overtime hours in the outbound department was partly substantiated: one of the two sites supports the claim. Punch items numbering 0 remained at Ellesmere Port when pilot cell go-live was signed on 2022-02-13. Berenice Straatman would re-buy on the basis of yes, subject to a commercial reset on overage, having first complained that the roadmap commitments given during the sales cycle slipped a year. Although Ellesmere Port treats it as outside anyone's control, punch-list closure review still moved to 2022-07-27 for customer middleware team could not release the interface test window. Della Fontanez-Ruck, speaking as warehouse systems manager, said that the analytics module is priced separately and was never budgeted and pressed the point twice before the call ended. Progress at Ellesmere Port shows site survey and slotting scan signed off on 2021-04-19 against 2021-02-14 as planned. Asked what worked, Della Fontanez-Ruck said that the account team escalated without being chased; asked what did not, the answer was that the analytics module is priced separately and was never budgeted. Wrexham closed wms interface build and message mapping on 2021-05-16 and the certificate records 0 open items. Lise Vandermolen, introduced by Pennarth Industrial as a willing reference, nonetheless complained that support handoffs across time zones lose context between shifts. Punch items numbering 0 remained at Wrexham when pilot cell go-live

was signed on 2021-09-03. Idunn Haraldsen volunteered that robots tolerated the mixed pallet and tote presentations on the floor, and the claim on reduction in temporary labour headcount at peak was not substantiated: the interviewee described it as an impression.

Fleet size across the estate stands at 78 robots spread over Wrexham, Ellesmere Port, and the distribution is uneven enough that account health is really the health of two or three buildings. The spread across sites is what matters: a strong building can carry a weak one through a renewal cycle, and a weak building can generate the ticket volume and credit exposure that later frames the commercial negotiation. Juniper Halloran-Voss holds the account on the company side and has done so without interruption, which is more continuity than several other files in this pack show. Notes taken on 2026-04-08 record Magdalena Ruiz-Hollander praising that the account team escalated without being chased beside the complaint that overage billing surprised the site in two consecutive quarters. Deployment records at Ellesmere Port leave acceptance test protocol execution closing on 2022-04-01, status closed. Whatever the company's materials claim, Magdalena Ruiz-Hollander described the position on order cycle time from release to pack-out as partly substantiated: one of the two sites supports the claim. Ellesmere Port is one of the few sites where network and radio-frequency readiness review ran to plan, closing on 2021-05-30. Asked what worked, Idunn Haraldsen said that robots tolerated the mixed pallet and tote presentations on the floor; asked what did not, the answer was that battery swap procedure is awkward for the night crew. Turning to Ellesmere Port, the tracker puts integration test in the customer sandbox at 2021-11-26 where the plan said 2021-09-20, the gap being labour agreement consultation with the site works council. Talking to Berenice Straatman for 36 minutes by video call with a follow-up email produced the claim that cost per outbound line improved by 21 per cent, unverifiable: the customer declined to release underlying data. Wrexham (PEN-01) met punch-list closure review on 2022-02-06 without a variation to plan. Because Cyriel Vandenbroucke sees the fleet every shift, the first question put was about dock-to-stock hours on inbound receipts, answered at 27 per cent. Deployment records at Wrexham leave network and radio-frequency readiness review closing on 2021-04-12, status closed. Notes taken on 2026-04-08 record Magdalena Ruiz-Hollander praising that the account team escalated without being chased beside the complaint that overage billing surprised the site in two consecutive quarters.

Metered usage against invoices at Pennarth Industrial

Metered volume at Pennarth Industrial exceeded entitlement in 8 of the 18 periods examined, and the excess is seasonal rather than structural, peaking in 2021-12 at 157,762 against an entitlement of 113,599. Invoices follow the meter with two exceptions, both of which were traced to an entitlement figure keyed from a superseded order form. Several periods carry credit notes, 6 in total, and the reasons stated on those notes do not always match the reasons the site gave us. Credits reaching Pennarth Industrial in the March cycle came to $1,092, and the balance remains as disputed; invoice held pending a joint meter review. Billing for April charged Pennarth Industrial $49,920 on a metered volume of 109,907, and the exception was no exception raised. Meter and invoice diverge in November at Pennarth Industrial by $320, which the parties treat as resolved: meter reconciliation accepted by both parties. Billing for February charged Pennarth Industrial $49,920 on a metered volume of 80,272, and the exception was partially credited; the balance remains contested. Pennarth Industrial was credited in May with $0, against an invoiced amount of $55,359. Pennarth Industrial used in January only 105,510 of an entitled 113,599, and no credit was sought in that cycle. Pennarth Industrial carried overage in March of 17,607 picks, which the invoice reflects and the site disputes on definition rather than count. Consumption recorded for Pennarth Industrial in November ran to 143,325 picks, 29,726 of them above entitlement. Overage in July at Pennarth Industrial came to 0 picks, entitlement having been fixed at signature. Pennarth Industrial's ledger for June shows overage of 0 picks, credits of $0 and a variance of $55, no exception raised. Pennarth Industrial in March shows entitlement of 113,599 against consumption of 131,206, recorded as no exception raised.

Reconciliation was performed period by period rather than on annual totals, because an annual view nets an overbilling in one month against an underbilling in another and hides both. Two directional errors were found: the site was overbilled where entitlement was understated, and underbilled in a period where the console under-recorded picks during a firmware regression. The under-billing was news to both parties, which says something about the strength of the meter-to-invoice control. Billing for December charged Pennarth Industrial $54,646 on a metered volume of 131,776, and the exception was no exception raised. Pennarth Industrial carried overage in May of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Reconciling July leaves Pennarth Industrial with a variance of $-333, partially credited; the balance remains contested. Reconciling February leaves Pennarth Industrial with a variance of

$-333, partially credited; the balance remains contested. Meter and invoice diverge in August at Pennarth Industrial by $117, which the parties treat as no exception raised. Reconciling April leaves Pennarth Industrial with a variance of $-277, no exception raised. Overage in October at Pennarth Industrial came to 44,740 picks, entitlement having been fixed at signature. Pennarth Industrial in September shows entitlement of 113,599 against consumption of 120,793, recorded as no exception raised. Pennarth Industrial's position for August was settled as no exception raised, on an invoice of $52,775. Credits reaching Pennarth Industrial in the September cycle came to $0, and the balance remains as no exception raised. Pennarth Industrial used in December only 131,776 of an entitled 113,599, and no credit was sought in that cycle.

Grainne Mulqueeny treats the overage mechanic as the main commercial grievance and said plainly that the site cannot forecast a bill it cannot model. The entitlement definition is the root of that complaint, since a re-pick after an exception counts against entitlement under the order form and does not count in the site's own operational reporting. The language does not settle the question, which is a contract-quality observation as much as a customer-relationship one, and it has been flagged to the legal workstream with the clause reference. Meter and invoice diverge in February at Pennarth Industrial by $-333, which the parties treat as partially credited; the balance remains contested. Billing for May charged Pennarth Industrial $49,920 on a metered volume of 101,387, and the exception was credited in the following billing cycle. Credits reaching Pennarth Industrial in the March cycle came to $0, and the balance remains as no exception raised. Pennarth Industrial was credited in January with $2,372, against an invoiced amount of $49,920. Reconciling February leaves Pennarth Industrial with a variance of $-277, credited in the following billing cycle. Pennarth Industrial metered December at 157,762 picks against an entitlement of 113,599, and was invoiced $61,402. Pennarth Industrial metered July at 114,166 picks against an entitlement of 113,599, and was invoiced $50,067. Pennarth Industrial carried overage in July of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Pennarth Industrial used in August only 124,580 of an entitled 113,599, and no credit was sought in that cycle. Reconciling March leaves Pennarth Industrial with a variance of $182, no exception raised. Meter and invoice diverge in August at Pennarth Industrial by $277, which the parties treat as escalated to the account executive, unresolved at fieldwork close.

Concentration of the customer base seen from the Pennarth Industrial file

Thornquist Parcel is the largest single relationship in the base at 14.5 per cent of recurring value, and the top three accounts together carry 35.3 per cent. Pennarth Industrial ranks 12 on the same measure, which is why its renewal is treated as a value-relevant event rather than as routine account management. Shares are expressed against the customer base only, and no consolidated figure is asserted here because the reconciliation to management's own reporting sits with the finance workstream. Pennarth Industrial carried overage in May of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Ellery Vandersloot sponsors the Arborlight Home deployment, which covers 8 sites in omnichannel home goods retail. Pennarth Industrial used in January only 77,423 of an entitled 113,599, and no credit was sought in that cycle. Set beside Palo Verde Electronics, which runs 2 sites at 57 robots, this account is the better documented of the two. Consumption recorded for Pennarth Industrial in June ran to 101,860 picks, 0 of them above entitlement. Comparison with Vantageport Terminals is instructive, since it prices on hardware plus software, with a spares pool subscription and pays on net sixty. Pennarth Industrial's position for October was settled as closed with no adjustment after evidence review, on an invoice of $54,045. Halcyon Medical Supply began on 2021-05-24 and its structure is described as auto-renew annually; expansion order signed in the third year. Console records for October at Pennarth Industrial show 129,466 picks, and the invoice followed at $54,045. Whatever this account decides, Arborlight Home reaches its own renewal on 2026-11-22 with expansion signed for a second wave of robots. Pennarth Industrial was credited in November with $0, against an invoiced amount of $57,648.

Loss sensitivity was run rather than described, because the interesting question is not the share of the largest account but what the second-largest becomes if the largest goes. This is a concentrated base, not a dependent one: losing the top account leaves a base whose new leader is close behind, so the shape does not collapse into a single-customer dependency. Contract expiry timing is the aggravating factor, since three of the five largest contracts reach a renewal decision within four quarters of each other. Reconciling March leaves Pennarth Industrial with a variance of $168, disputed; invoice held pending a joint meter review. Crestmoor Office Products renews on 2026-06-29, and Marcus Adeyinka-Blau owns that conversation on the company side. Meter and invoice diverge in November at Pennarth Industrial by $320, which the parties treat as resolved: meter reconciliation accepted by both parties. The nearest

comparable on fleet density is Grindelwald Precision, priced on per-robot subscription with an on-site engineer allocation. Pennarth Industrial was credited in August with $0, against an invoiced amount of $52,775. The nearest comparable on fleet density is Ironbark Building Materials, priced on hardware purchase plus software subscription. Billing for October charged Pennarth Industrial $54,045 on a metered volume of 129,466, and the exception was closed with no adjustment after evidence review. Set beside Brackenhill Pet Nutrition, which runs 3 sites at 116 robots, this account is the better documented of the two. Pennarth Industrial metered June at 95,233 picks against an entitlement of 113,599, and was invoiced $49,920. Among the peer accounts it is Halcyon Medical Supply where overage dispute unresolved through two escalation levels. Reconciling February leaves Pennarth Industrial with a variance of $-277, credited in the following billing cycle.

Solstice Renewables requires separate mention because it is an affiliate of an existing shareholder, and its pricing is stated in the order form as a most-favoured rate. Identification comes before judgement on related-party revenue, and the amount involved is small enough that its exclusion would not change the shape of the base materially. The pricing clause has wider consequences, because on its face it could be read to constrain pricing elsewhere, and that reading has been referred to the legal workstream with the clause identified. Lindenvale Grocery sits in member-owned grocery wholesale on per-robot subscription, cooperative member pricing schedule, with renewal falling on 2028-02-14 and notice due by 2027-11-16. Reconciling January leaves Pennarth Industrial with a variance of $55, no exception raised. Kestrel Pharma renews on 2026-06-30, and Aleksy Wisniewski owns that conversation on the company side. Pennarth Industrial was credited in December with $0, against an invoiced amount of $54,646. Caldera Cold Chain began on 2022-01-10 and its structure is described as fixed term with a single five-year extension option. Pennarth Industrial carried overage in March of 17,607 picks, which the invoice reflects and the site disputes on definition rather than count. Zephyrine Beauty Brands deploys 52 robots across 2 sites and is banded at 0.6m to 0.8m. Pennarth Industrial carried overage in July of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Crestmoor Office Products deploys 142 robots across 4 sites and is banded at 0.8m to 1.0m. Console records for February at Pennarth Industrial show 80,272 picks, and the invoice followed at $49,920. Meridian Fulfilment pays on net forty-five, quarterly in advance, a longer cycle than most of the base and one negotiated at signature.

Acceptance certificate evidence: Pennarth Industrial

Pennarth Industrial puts on the record that certificates were signed by operating staff rather than by procurement at every site, and the schedule below is the evidence for it, drawn from 2 records rather than from a summary prepared for us. Pennarth Industrial's file produces one more finding: one final certificate was withheld rather than refused, which is a different position. Pennarth Industrial is scheduled at record level rather than in summary, because an account-level average at Pennarth Industrial conceals the site-to-site variation that decides how this relationship behaves at renewal. At Wrexham the gate for pilot cell go-live was planned for 2021-08-18 and closed on 2021-09-03, the delay being customer sponsor on medical leave, sign-off authority unclear. Talking to Berenice Straatman for 36 minutes by video call with a follow-up email produced the claim that cost per outbound line improved by 21 per cent, unverifiable: the customer declined to release underlying data. Ellesmere Port lost time because seasonal peak freeze: no cutovers permitted during the retail quarter, and site survey and slotting scan moved to 2021-04-19 from a planned 2021-02-14. Praise from Magdalena Ruiz-Hollander took the form that the account team escalated without being chased, and it came without prompting. Ellesmere Port is one of the few sites where network and radio-frequency readiness review ran to plan, closing on 2021-05-30. Because Idunn Haraldsen sees the fleet every shift, the first question put was about reduction in temporary labour headcount at peak, answered at 37 per cent. Nothing needed explaining at Wrexham for punch-list closure review, which was reached on 2022-02-06 as planned. Petra Lindenbaum-Cruz, speaking as distribution centre operations manager, said that reporting cannot be reconciled against the customer's own labour standard and pressed the point twice before the call ended.

The Pennarth Industrial workpaper is qualified in two respects: dates are certificate dates, which occasionally postdate the underlying event, and the schedule excludes amounts withheld for reasons unrelated to acceptance. Grainne Mulqueeny also observed, when the schedule was put to the customer, that withheld amounts were agreed with the site rather than taken from the company ledger, which is consistent with what the documents show without being proved by them. Teodor Vaszary accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Clementine Ashdown, introduced by Pennarth Industrial as a willing reference,

nonetheless complained that traffic management degrades when aisle congestion exceeds design load. Punch items numbering 9 remained at Ellesmere Port when conditional acceptance certificate was signed on 2022-05-05. Talking to Berenice Straatman for 36 minutes by video call with a follow-up email produced the claim that cost per outbound line improved by 21 per cent, unverifiable: the customer declined to release underlying data. Certificates held at Ellesmere Port put acceptance test protocol execution on 2022-04-01 and explain the 4 days by seasonal peak freeze: no cutovers permitted during the retail quarter. Asked what worked, Magdalena Ruiz-Hollander said that the account team escalated without being chased; asked what did not, the answer was that overage billing surprised the site in two consecutive quarters. Deployment records at Ellesmere Port leave conditional acceptance certificate closing on 2022-05-05, status conditional acceptance issued. Clementine Ashdown was reached by telephone and gave a re-buy answer of yes, without qualification, which sits oddly beside the praise that firmware rollbacks were quick when a release misbehaved. Certificates held at Ellesmere Port put site survey and slotting scan on 2021-04-19 and explain the 64 days by seasonal peak freeze: no cutovers permitted during the retail quarter.

Schedule 3.2 - Pennarth Industrial: acceptance certificates and amounts withheld

Site	Conditional acceptance date	Final acceptance date	Punch items at conditional	Punch items open	Withheld amount
Wrexham (PEN-01)	2021-12-24	2022-03-21	5	0	nil
Ellesmere Port (PEN-02)	2022-05-05	2022-08-12	9	0	nil

Deployment sequence and go-live history at Pennarth Industrial

Wrexham (PEN-01) took the first site survey on 2021-01-08 against a plan of 2021-01-08, and the 0-day difference was attributed to no slip: milestone met inside the planned window. Sequencing thereafter followed the standard pattern: radio-frequency readiness, interface build against the host warehouse-management system, sandbox integration testing, then a pilot cell before any protocol was executed. Wrexham (PEN-01) reached pilot go-live on 2021-09-03, which is outside the window the order form contemplated. Pennarth Industrial let us check the published reference case study against the customer's own copy, and it shows what was claimed to the market. Turning to Ellesmere Port, the tracker puts conditional acceptance certificate at 2022-05-05 where the plan said 2022-04-30, the gap being charger electrical permit issued late by the local authority. Pennarth Industrial let us check the published reference case study against the customer's own copy, and it shows what was claimed to the market. Wrexham logged integration test in the customer sandbox on 2021-07-27; the customer attributes the slip to supplier lead time on the conveyor tie-in guard package. Pennarth Industrial was asked for the conditional acceptance certificate precisely because it bears on which items were carried forward as punch. Deployment records at Wrexham leave pilot cell go-live closing on 2021-09-03, status closed. Pennarth Industrial let us check the works council consultation note against the customer's own copy, and it shows the labour constraints on further automation. Where the plan assumed a clean run, Ellesmere Port instead took 5 extra days over conditional acceptance certificate, for charger electrical permit issued late by the local authority. Pennarth Industrial relies on the joint escalation minutes for its own account of the position, and it shows what was agreed when the relationship was under strain. Progress at Ellesmere Port shows site survey and slotting scan signed off on 2021-04-19 against 2021-02-14 as planned. Pennarth Industrial let us check the published reference case study against the customer's own copy, and it shows what was claimed to the market.

Interface work is where this deployment differed from the company's published sequence, because Pennarth Industrial runs a host system whose release calendar it does not control and would not move. Message design absorbed most of the elapsed time, and two of the mapping rounds were driven by customer master data on slot dimensions rather than by anything the company controlled. Responsibility for the delay is contested there, and it is examined against the documents later in this file rather than settled here. The site labour standard sits on the Pennarth Industrial file and is the source for the baseline against which benefits are claimed. Once Ellesmere Port had dealt with wireless survey failed in the high-bay aisles, access points re-sited, the site completed pilot cell go-live on 2022-02-13. Reading the amended order form for Pennarth Industrial is what establishes nothing, because it was never produced. Acceptance slipped 4 days at Ellesmere Port before acceptance test protocol execution was recorded as closed. The pilot charter for Pennarth Industrial was produced on request and shows what the parties agreed the pilot had to prove. Turning to Ellesmere Port, the tracker puts integration test in the customer sandbox at 2021-11-26 where the plan said 2021-09-20, the gap being labour agreement consultation with the site works council. The joint escalation minutes at Pennarth Industrial carries more

weight than any summary, since it shows what was agreed when the relationship was under strain. Deployment records at Wrexham leave acceptance test protocol execution closing on 2021-11-04, status closed. The support ticket export held by Pennarth Industrial is the document that decides the point, relied upon for how each incident was classified for severity. Ellesmere Port logged pilot cell go-live on 2022-02-13; the customer attributes the slip to wireless survey failed in the high-bay aisles, access points re-sited. The site labour standard in the Pennarth Industrial bundle is cited throughout this review as the evidence for the baseline against which benefits are claimed.

Teodor Vaszary described the deployment as slower than promised but not badly run, and offered the charger electrical works as an example of a delay the site itself caused and did not blame the vendor for. Company implementation notes read differently, recording the same period as awaiting customer readiness without naming the electrical permit, so the two records agree on the fact and disagree on the framing. Both readings are carried forward because the pattern recurs across several accounts in this pack and is itself a finding. Ellesmere Port lost time because facility readiness: mezzanine steel and charger bays incomplete, and wms interface build and message mapping moved to 2021-08-13 from a planned 2021-06-22. Reading the console meter extract for Pennarth Industrial is what establishes the volume the company recorded. At Ellesmere Port the gate for final acceptance certificate closed on 2022-08-12 inside the planned window, and no cause needs recording. The renewal correspondence file in the Pennarth Industrial bundle is cited throughout this review as the evidence for what each party put in writing about intent. At Ellesmere Port the gate for integration test in the customer sandbox was planned for 2021-09-20 and closed on 2021-11-26, the delay being labour agreement consultation with the site works council. Pennarth Industrial relies on the pilot charter for its own account of the position, and it shows what the parties agreed the pilot had to prove. At Ellesmere Port the gate for network and radio-frequency readiness review closed on 2021-05-30 inside the planned window, and no cause needs recording. Pennarth Industrial let us check the works council consultation note against the customer's own copy, and it shows the labour constraints on further automation. Acceptance slipped 5 days at Ellesmere Port before conditional acceptance certificate was recorded as conditional acceptance issued. The site labour standard in the Pennarth Industrial bundle is cited throughout this review as the evidence for the baseline against which benefits are claimed. Ellesmere Port lost time because labour agreement consultation with the site works council, and integration test in the customer sandbox moved to 2021-11-26 from a planned 2021-09-20.

Pennarth Industrial - punch-list evidence

The Ellesmere Port operation records that punch counts fell sharply once an on-site engineer was assigned, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. Pennarth Industrial supports a related reading: several open items are functional rather than cosmetic and touch the host interface. The Ellesmere Port operation is scheduled at record level rather than in summary, because an account-level average at Pennarth Industrial conceals the site-to-site variation that decides how this relationship behaves at renewal. The conditional acceptance certificate in the Pennarth Industrial bundle is cited throughout this review as the evidence for which items were carried forward as punch. Wrexham (PEN-01) reached network and radio-frequency readiness review on 2021-04-12, status closed, with 0 punch items open at that gate. The conditional acceptance certificate held by Pennarth Industrial is the document that decides the point, relied upon for which items were carried forward as punch. Ellesmere Port logged wms interface build and message mapping on 2021-08-13; the customer attributes the slip to facility readiness: mezzanine steel and charger bays incomplete. The site labour standard for Pennarth Industrial was produced on request and shows the baseline against which benefits are claimed. Wrexham (PEN-01) met punch-list closure review on 2022-02-06 without a variation to plan. The joint escalation minutes at Pennarth Industrial carries more weight than any summary, since it shows what was agreed when the relationship was under strain. The slip of 4 days at Ellesmere Port on acceptance test protocol execution is the one the customer raises first, and it traces to seasonal peak freeze: no cutovers permitted during the retail quarter.

Pennarth Industrial's file carries two qualifications: the schedule shows items open at fieldwork date, not items ever raised, and closure evidence was sighted for some items and asserted for others. The Pennarth Industrial workpaper also observed, when the schedule was put to the customer, that punch items are concentrated at the gates closest to final acceptance, which is consistent with what the documents show without being proved by them. Grainne Mulqueeny accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. The support ticket export for Pennarth Industrial was produced on request and shows how each

incident was classified for severity. Where the plan assumed a clean run, Ellesmere Port instead took 5 extra days over conditional acceptance certificate, for charger electrical permit issued late by the local authority. The console meter extract sits on the Pennarth Industrial file and is the source for the volume the company recorded. Deployment records at Wrexham leave final acceptance certificate closing on 2022-03-21, status final acceptance granted. The conditional acceptance certificate in the Pennarth Industrial bundle is cited throughout this review as the evidence for which items were carried forward as punch. Deployment records at Wrexham leave network and radio-frequency readiness review closing on 2021-04-12, status closed. The support ticket export at Pennarth Industrial carries more weight than any summary, since it shows how each incident was classified for severity. Punch items numbering 0 remained at Ellesmere Port when wms interface build and message mapping was signed on 2021-08-13.

Schedule 3.3 - Pennarth Industrial: punch-list position at each acceptance gate

Site	Milestone	Actual completion	Open punch items	Status	Days beyond plan
Wrexham (PEN-01)	Conditional acceptance certificate	2021-12-24	5	conditional acceptance issued	14
Wrexham (PEN-01)	Punch-list closure review	2022-02-06	5	punch list closed	0
Wrexham (PEN-01)	Final acceptance certificate	2022-03-21	0	final acceptance granted	22

Support attainment evidence: Pennarth Industrial

The Pennarth Industrial estate puts on the record that one site accounts for a disproportionate share of severity-one events, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. This Pennarth Industrial review supports a related reading: ticket volume tracks fleet size loosely and shift count closely. The Ellesmere Port operation carries the detail at record level deliberately, because an account-level average at Pennarth Industrial conceals the site-to-site variation that decides how this relationship behaves at renewal. Attainment figures from Ellesmere Port show 97 per cent on response against 82 per cent on resolution. Wrexham (PEN-01) met response targets 87 per cent of the time, and resolution targets rather less often. Ellesmere Port (PEN-02) is where lift-plate calibration drift recurs, and the quarter to 2021-Q4 shows 2 severity-one events. Although the console reported no outage at Wrexham, the site still raised 33 tickets in 2021-Q2. Ellesmere Port keeps returning to one theme, fleet console latency during shift change, and it survived into 2022-Q4. Ellesmere Port (PEN-02) is where lift-plate calibration drift recurs, and the quarter to 2021-Q4 shows 2 severity-one events. Reviewing the 2023-Q2 quarter, Ellesmere Port shows 31 tickets and a recurring theme of traffic deadlock at an aisle intersection. Ellesmere Port (PEN-02) shows traffic deadlock at an aisle intersection as the complaint theme, on 31 tickets in 2022-Q3.

The Ellesmere Port building must be read subject to two limits: quarters are contract quarters and do not align to calendar quarters at every site, and credits shown are those issued, not those the site believes were due. Diligence on Pennarth Industrial also observed, when the schedule was put to the customer, that credits were issued in the quarters with the weakest resolution attainment, which is consistent with what the documents show without being proved by them. The Wrexham site accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Ticket volume reached 26 at Ellesmere Port in 2022-Q1, of which 1 were classified at the highest severity. Ticket volume reached 30 at Wrexham in 2023-Q3, of which 2 were classified at the highest severity. Wrexham triggered credits of $4,433 following resolution attainment of 64 per cent. Although the console reported no outage at Wrexham, the site still raised 28 tickets in 2022-Q4. Wrexham (PEN-01) is where wireless roaming loss in the cold zone recurs, and the quarter to 2021-Q2 shows 1 severity-one events. Recurring at Wrexham through that quarter was traffic deadlock at an aisle intersection, across 28 tickets. Wrexham triggered credits of $4,433 following resolution attainment of 64 per cent. Ticket volume reached 38 at Wrexham in 2022-Q3, of which 0 were classified at the highest severity.

Schedule 3.4 - Pennarth Industrial: support attainment against contractual commitments

Site	Period	Tickets	Severity-1	Response met (%)	Resolution met (%)	Credits issued
Wrexham (PEN-01)	2021-Q2	33	1	90	91	nil
Wrexham (PEN-01)	2021-Q3	21	1	98	64	$4,433
Wrexham	2021-Q4	28	1	94	90	nil

Site	Period	Tickets	Severity-1	Response met (%)	Resolution met (%)	Credits issued
(PEN-01)						

Escalation chronology supporting the Pennarth Industrial review

The Ellesmere Port building evidences that credits followed escalation in every case and never preceded it, and the schedule below is the evidence for it, drawn from 3 records rather than from a summary prepared for us. Diligence on Pennarth Industrial supports a related reading: the contractual hook is missing from more escalations than it is present in. The Pennarth Industrial bundle chose this grain deliberately, because an account-level average at Pennarth Industrial conceals the site-to-site variation that decides how this relationship behaves at renewal. Pennarth Industrial was credited in January with $2,372, against an invoiced amount of $49,920. Ellesmere Port reached 93 per cent on response and 62 per cent on resolution, and credits of $6,270 followed. Credits reaching Pennarth Industrial in the May cycle came to $5,524, and the balance remains as credited in the following billing cycle. Ellesmere Port produced 0 severity-one events in 2021-Q3, and resolution attainment of 90 per cent triggered credits of $0. Pennarth Industrial used in March only 115,113 of an entitled 113,599, and no credit was sought in that cycle. Although the console reported no outage at Ellesmere Port, the site still raised 31 tickets in 2023-Q2. Pennarth Industrial's ledger for February shows overage of 0 picks, credits of $0 and a variance of $166, no exception raised. Attainment figures from Ellesmere Port show 99 per cent on response against 70 per cent on resolution.

The Wrexham site comes with two limits on its use: an item shown as still open was open at fieldwork close, and the chronology is assembled from three record sets and may omit informal contacts. Pennarth Industrial management also observed, when the schedule was put to the customer, that the sponsor rather than procurement raised most of the operational escalations, which is consistent with what the documents show without being proved by them. Fieldwork at Pennarth Industrial accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Attainment figures from Ellesmere Port show 98 per cent on response against 90 per cent on resolution. Pennarth Industrial in December shows entitlement of 113,599 against consumption of 140,438, recorded as credited in the following billing cycle. Wrexham (PEN-01) met response targets 98 per cent of the time, and resolution targets rather less often. Pennarth Industrial in February shows entitlement of 113,599 against consumption of 76,202, recorded as no exception raised. Ellesmere Port produced 1 severity-one events in 2022-Q1, and resolution attainment of 78 per cent triggered credits of $2,758. Pennarth Industrial in July shows entitlement of 113,599 against consumption of 97,127, recorded as partially credited; the balance remains contested. Attainment figures from Ellesmere Port show 98 per cent on response against 75 per cent on resolution. Overage in December at Pennarth Industrial came to 18,177 picks, entitlement having been fixed at signature.

Schedule 3.5 - Pennarth Industrial: escalation chronology and contractual hooks

Date	Event	Raised with	Contractual hook	Outcome
2021-11-12	severity-one raised: fleet halted on one floor	the customer success director	no contractual hook cited	closed with a credit
2021-12-13	customer wrote to dispute an invoice	the company vice president of services	service level schedule, clause on resolution credits	closed with a credit
2021-12-30	joint meter review held	the company account executive	acceptance schedule, cure period	closed after a firmware release

Site and fleet inventory: Pennarth Industrial

Grainne Mulqueeny's team evidences that one site carries a fleet dense enough to be a reference in its own right, and the schedule below is the evidence for it, drawn from 2 records rather than from a summary prepared for us. The Ellesmere Port operation supports a related reading: site codes here follow the company's own numbering so the schedule can be tied back. The Wrexham team settled on this level of detail for a reason, because an account-level average at Pennarth Industrial conceals the site-to-site variation that decides how this relationship behaves at renewal. Ellesmere Port reached 85 per cent on response and 78 per cent on resolution, and credits of $2,758 followed. Turning to Wrexham, the tracker puts network and radio-frequency readiness review at 2021-04-12 where the plan said 2021-02-08, the gap being customer master-data cleanup for slot dimensions ran long. Ellesmere Port keeps returning to one theme, charger handshake failures on one dock, and it survived into 2021-Q2. Acceptance slipped 4 days at Ellesmere Port before acceptance test protocol execution was recorded as closed. Ellesmere Port produced 1 severity-one events in 2022-Q1,

and resolution attainment of 78 per cent triggered credits of $2,758. Wrexham (PEN-01) cleared final acceptance certificate on 2022-03-21 against a plan of 2022-02-27, a difference of 22 days recorded against wireless survey failed in the high-bay aisles, access points re-sited. Although the console reported no outage at Wrexham, the site still raised 28 tickets in 2022-Q4. Turning to Ellesmere Port, the tracker puts integration test in the customer sandbox at 2021-11-26 where the plan said 2021-09-20, the gap being labour agreement consultation with the site works council.

Pennarth Industrial needs two caveats stated: a decommissioned site is retained in the schedule rather than deleted, so the footprint is not overstated, and go-live is recorded as first productive pick, not as the date of the first robot on site. Pennarth Industrial's file also observed, when the schedule was put to the customer, that the estate was built out over more waves than the company's rollout narrative suggests, which is consistent with what the documents show without being proved by them. The Pennarth Industrial workpaper accepted the schedule as a fair summary of the underlying records, subject to those qualifications and to the open items listed in this review. Ellesmere Port keeps returning to one theme, charger handshake failures on one dock, and it survived into 2021-Q2. Wrexham logged integration test in the customer sandbox on 2021-07-27; the customer attributes the slip to supplier lead time on the conveyor tie-in guard package. Reviewing the 2023-Q2 quarter, Ellesmere Port shows 31 tickets and a recurring theme of traffic deadlock at an aisle intersection. Progress at Ellesmere Port shows acceptance test protocol execution signed off on 2022-04-01 against 2022-03-28 as planned. Recurring at Wrexham through that quarter was interface message backlog after a host restart, across 30 tickets. Wrexham lost time because supplier lead time on the conveyor tie-in guard package, and conditional acceptance certificate moved to 2021-12-24 from a planned 2021-12-10. Ellesmere Port produced 0 severity-one events in 2021-Q3, and resolution attainment of 90 per cent triggered credits of $0. Wrexham (PEN-01) cleared network and radio-frequency readiness review on 2021-04-12 against a plan of 2021-02-08, a difference of 63 days recorded against customer master-data cleanup for slot dimensions ran long.

Schedule 3.6 - Pennarth Industrial: site fleet inventory and shift coverage

Site code	City	Robots	Shift coverage	Charger bays	Go-live
PEN-01	Wrexham	37	two shifts, seven days	4	2021-02-01
PEN-02	Ellesmere Port	41	one shift, five days	12	2021-06-01

Quantified benefit claims at Pennarth Industrial and how far they stand up

Cyriel Vandenbroucke put the improvement in dock-to-stock hours on inbound receipts at 27 per cent, and that figure was tested rather than recorded, because a benefit claim from a reference is a marketing input until someone reconciles it to a report. Supporting data was offered rarely: 1 of 5 claims were substantiated to the point where the direction and rough magnitude both hold. No claim was contradicted outright, which is the outcome that would matter most to an investor reading the reference pack. Credits reaching Pennarth Industrial in the March cycle came to $0, and the balance remains as no exception raised. Cyriel Vandenbroucke treats the relationship as one where charge management stopped being a supervisor problem after tuning, and the dock-to-stock hours on inbound receipts figure offered was 27 per cent. Meter and invoice diverge in February at Pennarth Industrial by $-333, which the parties treat as partially credited; the balance remains contested. Clementine Ashdown was reached by telephone and gave a re-buy answer of yes, without qualification, which sits oddly beside the praise that firmware rollbacks were quick when a release misbehaved. Pennarth Industrial used in March only 115,113 of an entitled 113,599, and no credit was sought in that cycle. Berenice Straatman would re-buy on the basis of yes, subject to a commercial reset on overage, having first complained that the roadmap commitments given during the sales cycle slipped a year. Credits reaching Pennarth Industrial in the May cycle came to $5,524, and the balance remains as credited in the following billing cycle. Nkechi Adaeze-Fowler treats the relationship as one where safety record was clean across two shifts and no lost-time incidents, and the inventory record accuracy after cycle counting figure offered was 20 per cent. Pennarth Industrial's position for January was settled as disputed; invoice held pending a joint meter review, on an invoice of $49,920. Asked what worked, Cyriel Vandenbroucke said that charge management stopped being a supervisor problem after tuning; asked what did not, the answer was that the analytics module is priced separately and was never budgeted. Billing for January charged Pennarth Industrial $49,920 on a metered volume of 111,209, and the exception was disputed; invoice held pending a joint meter review.

Baseline choice is the recurring weakness in these claims, and it is not usually deliberate. Baselines are chosen after the fact, often one that predates a network redesign or a labour agreement change, so an improvement attributed to the

fleet includes gains the fleet did not produce. We re-based two claims onto comparable periods, and both fell materially once the comparison period was moved to a like-for-like quarter. Obinna Chukwuemeka volunteered that implementation engineers were candid about what the pilot would not prove, and the claim on cost per outbound line was substantiated: the site produced labour reports for the comparison periods. Meter and invoice diverge in May at Pennarth Industrial by $-369, which the parties treat as written off as a goodwill concession. Clementine Ashdown raised the complaint that traffic management degrades when aisle congestion exceeds design load, and put the improvement in dock-to-stock hours on inbound receipts at 23 per cent. Pennarth Industrial was credited in June with $0, against an invoiced amount of $49,920. Nkechi Adaeze-Fowler treats the relationship as one where safety record was clean across two shifts and no lost-time incidents, and the inventory record accuracy after cycle counting figure offered was 20 per cent. Consumption recorded for Pennarth Industrial in March ran to 115,113 picks, 1,514 of them above entitlement. Idunn Haraldsen volunteered that robots tolerated the mixed pallet and tote presentations on the floor, and the claim on reduction in temporary labour headcount at peak was not substantiated: the interviewee described it as an impression. Pennarth Industrial carried overage in December of 44,163 picks, which the invoice reflects and the site disputes on definition rather than count. Because Petra Lindenbaum-Cruz sees the fleet every shift, the first question put was about overtime hours in the outbound department, answered at 20 per cent. Pennarth Industrial metered October at 129,466 picks against an entitlement of 113,599, and was invoiced $54,045. Praise from Nkechi Adaeze-Fowler took the form that safety record was clean across two shifts and no lost-time incidents, and it came without prompting.

Management's own case study on this account states a benefit that no interviewee repeated in the same terms, and the gap is not explained by rounding or by a different measurement window. Management was asked for the supporting model and received a summary rather than a derivation, so the claim is treated as unverified in this file. We allege nothing beyond an evidence gap, and the open request is carried into the outstanding-items list at the end of this review. Della Fontanez-Ruck volunteered that the account team escalated without being chased, and the claim on overtime hours in the outbound department was partly substantiated: one of the two sites supports the claim. Pennarth Industrial's position for June was settled as no exception raised, on an invoice of $49,920. Clementine Ashdown, speaking as procurement category manager, said that traffic management degrades when aisle congestion exceeds design load and pressed the point twice before the call ended. Pennarth Industrial metered February at 76,202 picks against an entitlement of 113,599, and was invoiced $49,920. Whatever the company's materials claim, Magdalena Ruiz-Hollander described the position on order cycle time from release to pack-out as partly substantiated: one of the two sites supports the claim. Pennarth Industrial carried overage in January of 0 picks, which the invoice reflects and the site disputes on definition rather than count. Talking to Berenice Straatman for 36 minutes by video call with a follow-up email produced the claim that cost per outbound line improved by 21 per cent, unverifiable: the customer declined to release underlying data. Billing for February charged Pennarth Industrial $49,920 on a metered volume of 80,272, and the exception was partially credited; the balance remains contested. Della Fontanez-Ruck volunteered that the account team escalated without being chased, and the claim on overtime hours in the outbound department was partly substantiated: one of the two sites supports the claim. Pennarth Industrial used in September only 120,793 of an entitled 113,599, and no credit was sought in that cycle. Obinna Chukwuemeka, the labour planning analyst, spoke on 2026-04-03 and praised that implementation engineers were candid about what the pilot would not prove, answering the re-buy question with yes, and has already sponsored an expansion.

Reference conversations held on Pennarth Industrial

Cyriel Vandenbroucke, the quality assurance manager, spoke for 28 minutes on 2026-03-30 and praised that charge management stopped being a supervisor problem after tuning. Nkechi Adaeze-Fowler was less generous and led with the complaint that the roadmap commitments given during the sales cycle slipped a year, a point raised without prompting and repeated later in the call. The re-buy answers do not line up: 3 of 5 interviewees said yes in some form, and the qualifications attached to those yes answers are where the useful information sits. Praise from Magdalena Ruiz-Hollander took the form that the account team escalated without being chased, and it came without prompting. Asked what worked, Della Fontanez-Ruck said that the account team escalated without being chased; asked what did not, the answer was that the analytics module is priced separately and was never budgeted. Cyriel Vandenbroucke, speaking as quality assurance manager, said that the analytics module is priced separately and was never budgeted and pressed the point twice before the call ended. Clementine Ashdown, introduced by Pennarth Industrial as a willing reference, nonetheless complained

that traffic management degrades when aisle congestion exceeds design load. Cyriel Vandenbroucke treats the relationship as one where charge management stopped being a supervisor problem after tuning, and the dock-to-stock hours on inbound receipts figure offered was 27 per cent. Because Idunn Haraldsen sees the fleet every shift, the first question put was about reduction in temporary labour headcount at peak, answered at 37 per cent. Talking to Lise Vandermolen for 30 minutes by on-site walkthrough produced the claim that inventory record accuracy after cycle counting improved by 9 per cent, unverifiable: the customer declined to release underlying data. Clementine Ashdown, introduced by Pennarth Industrial as a willing reference, nonetheless complained that traffic management degrades when aisle congestion exceeds design load. Clementine Ashdown raised the complaint that traffic management degrades when aisle congestion exceeds design load, and put the improvement in dock-to-stock hours on inbound receipts at 23 per cent. Asked what worked, Magdalena Ruiz-Hollander said that the account team escalated without being chased; asked what did not, the answer was that overage billing surprised the site in two consecutive quarters. Asked what worked, Cyriel Vandenbroucke said that charge management stopped being a supervisor problem after tuning; asked what did not, the answer was that the analytics module is priced separately and was never budgeted.

Della Fontanez-Ruck gave the answer that most complicates the company's reference narrative, stating on re-buy that the position is yes, but only for high-velocity zones while simultaneously praising that the account team escalated without being chased. Satisfaction and re-purchase intent diverge here, and in this account they diverge along the operations-procurement line rather than by site. One theme was raised by several interviewees, namely that the roadmap commitments given during the sales cycle slipped a year, which no interviewee described as resolved. Petra Lindenbaum-Cruz raised the complaint that reporting cannot be reconciled against the customer's own labour standard, and put the improvement in overtime hours in the outbound department at 20 per cent. Idunn Haraldsen gave 57 minutes on 2026-04-12 and made one quantified claim, on reduction in temporary labour headcount at peak, of 37 per cent. Notes taken on 2026-04-12 record Idunn Haraldsen praising that robots tolerated the mixed pallet and tote presentations on the floor beside the complaint that battery swap procedure is awkward for the night crew. Della Fontanez-Ruck, speaking as warehouse systems manager, said that the analytics module is priced separately and was never budgeted and pressed the point twice before the call ended. Cyriel Vandenbroucke treats the relationship as one where charge management stopped being a supervisor problem after tuning, and the dock-to-stock hours on inbound receipts figure offered was 27 per cent. Obinna Chukwuemeka, the labour planning analyst, spoke on 2026-04-03 and praised that implementation engineers were candid about what the pilot would not prove, answering the re-buy question with yes, and has already sponsored an expansion. Asked what worked, Berenice Straatman said that the fleet-orchestration console gave supervisors usable congestion data; asked what did not, the answer was that the roadmap commitments given during the sales cycle slipped a year. Petra Lindenbaum-Cruz, the distribution centre operations manager, spoke on 2026-03-05 and praised that the WMS interface behaved predictably once message mapping settled, answering the re-buy question with no comment; referred the question to procurement. Whatever the company's materials claim, Lise Vandermolen described the position on inventory record accuracy after cycle counting as unverifiable: the customer declined to release underlying data. Whatever the company's materials claim, Obinna Chukwuemeka described the position on cost per outbound line as substantiated: the site produced labour reports for the comparison periods. Praise from Magdalena Ruiz-Hollander took the form that the account team escalated without being chased, and it came without prompting.

Conversations were conducted from a fixed question set so that answers stay comparable between accounts, with follow-up questions asked only where an answer conflicted with a document already in hand. Interviewees saw none of the vendor material, which is why a contradiction with the published case study counts as evidence rather than as a prompted reaction. That kind of contradiction came up twice, and both are set out in the register below together with the material they contradict. Notes taken on 2026-04-12 record Idunn Haraldsen praising that robots tolerated the mixed pallet and tote presentations on the floor beside the complaint that battery swap procedure is awkward for the night crew. Notes taken on 2026-04-12 record Idunn Haraldsen praising that robots tolerated the mixed pallet and tote presentations on the floor beside the complaint that battery swap procedure is awkward for the night crew. Cyriel Vandenbroucke, speaking as quality assurance manager, said that the analytics module is priced separately and was never budgeted and pressed the point twice before the call ended. Petra Lindenbaum-Cruz was reached by telephone and gave a re-buy answer of no comment; referred the question to procurement, which sits oddly beside the praise that the WMS interface behaved predictably once message mapping settled. Della Fontanez-Ruck gave 50 minutes on 2026-03-05 and made one quantified claim, on overtime hours in the outbound department, of 39 per cent. Praise from Cyriel Vandenbroucke took

the form that charge management stopped being a supervisor problem after tuning, and it came without prompting. Talking to Lise Vandermolen for 30 minutes by on-site walkthrough produced the claim that inventory record accuracy after cycle counting improved by 9 per cent, unverifiable: the customer declined to release underlying data. Obinna Chukwuemeka gave 37 minutes on 2026-04-03 and made one quantified claim, on cost per outbound line, of 17 per cent. Della Fontanez-Ruck, speaking as warehouse systems manager, said that the analytics module is priced separately and was never budgeted and pressed the point twice before the call ended. Idunn Haraldsen volunteered that robots tolerated the mixed pallet and tote presentations on the floor, and the claim on reduction in temporary labour headcount at peak was not substantiated: the interviewee described it as an impression. Talking to Magdalena Ruiz-Hollander for 51 minutes by telephone produced the claim that order cycle time from release to pack-out improved by 24 per cent, partly substantiated: one of the two sites supports the claim.